aiPHeed Food Insecurity Forecast Report

Province Risk Level Trend

Area	All provinces (Cavite, Laguna, Quezon, Rizal, Batangas)
Period	2023 Q1 to 2027 Q1
Date prepared	18 October 2026
Prepared for	DA Region IV-A
Data	2027 Q1 lead-2 preview, issued 2026-10-01

The risk score ranges from 0 to 1. It is the estimated chance that the province's hunger-and-food-price score -- regional hunger from the SWS survey plus twice the gap between the province's food inflation and the five-province average -- is above its usual level, the median of the past published quarters. A higher score means this combined pressure is more likely to be above its usual level. It is not a measure of how many families lack food. This report uses three risk levels: Alert (0.60 and above), High (0.50 to below 0.60) and Not elevated (below 0.50).

Key Points

- In 2027 Q1, all five provinces are at the Alert level.
- The CALABARZON index, the mean of the five provinces, is 0.80 (Alert).
- Cavite has the highest risk score (0.82) and Batangas the lowest (0.71).
- Food prices are the main factor behind the risk in every province; they raise it in Cavite, Laguna, Quezon, and Rizal and lower it in Batangas.
- Compared with 2023 Q1, the risk score increased in all five provinces.
- The 2027 Q1 value is a two-quarter-ahead experimental preview issued 2026-10-01; its outcome is not yet known.

Risk at a Glance

As shown in Table 1, all five provinces are at the Alert level in 2027 Q1. The table lists the provinces from the highest risk score to the lowest.

Table 1

Risk Scores and Risk Levels by Province, 2027 Q1

Province	Risk score	Risk level	Change from 2026 Q4	Main factor
Cavite	0.82	Alert	Down 0.10	Food prices (73%)
Laguna	0.82	Alert	Down 0.09	Food prices (73%)
Quezon	0.82	Alert	Down 0.10	Food prices (73%)
Rizal	0.82	Alert	Down 0.10	Food prices (73%)
Batangas	0.71	Alert	Up 0.43	Food prices (79%)
CALABARZON (mean)	0.80	Alert	Up 0.01	Food prices (76%)

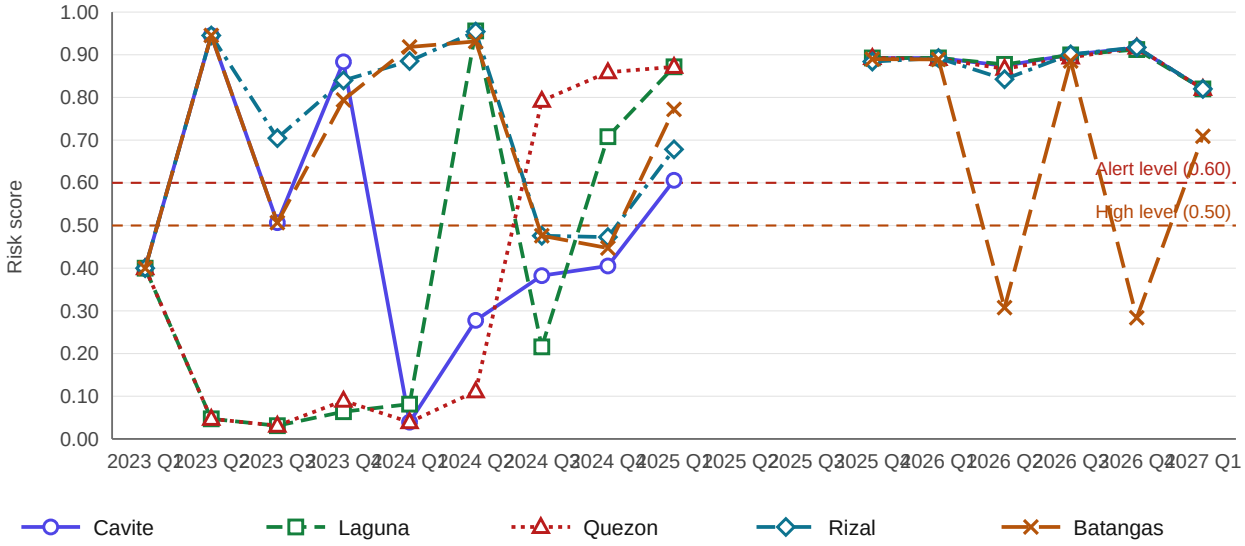
Note. Risk levels: Alert is 0.60 and above, High is 0.50 to below 0.60, and Not elevated is below 0.50. The main factor is the largest of the five factors behind each score. 2027 Q1: lead-2 preview, issued 2026-10-01.

How Risk Changed

As shown in Figure 1, between 2023 Q1 and 2027 Q1 the risk score increased in all five provinces. The largest change was in Cavite, where the score increased by 0.42 (from 0.40 to 0.82). The highest score in the period was 0.96, in Laguna in 2024 Q2.

Figure 1

Risk Scores for All Provinces, 2023 Q1 to 2027 Q1



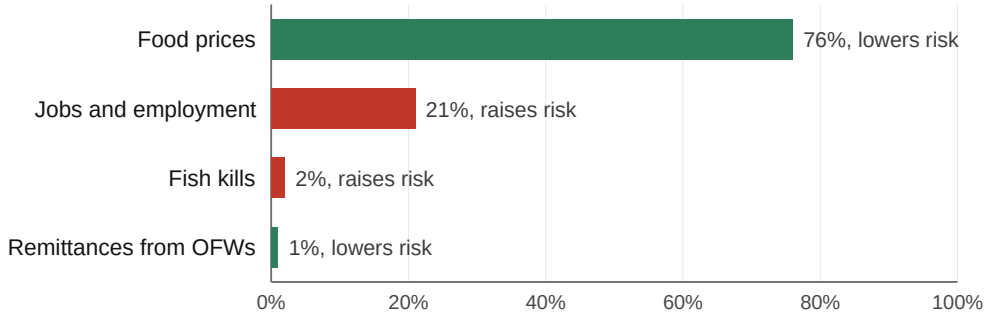
Note. Dashed lines mark the Alert level (0.60) and the High level (0.50). 2023 Q1 to 2026 Q3 are backtests on past quarters. 2026 Q4 is a forecast, lead 1, issued 2026-10-01. 2027 Q1 is a lead-2 preview, issued 2026-10-01. A gap in a line marks a quarter without a result. No estimate for 2025 Q2 and 2025 Q3: their SWS hunger rounds are recorded as supplementary, not as the approved (definition A) series, so these quarters have no approved label to estimate or score. This is missing data, not a low score.

What Is Driving the Risk

Figure 2 combines the factors behind the five provinces' risk scores in 2027 Q1, and Table 2 shows them province by province. In Cavite, Laguna, Quezon, and Rizal, food prices are the main factor, accounting for about 73% of the factors behind the score, and they raise the risk. In Batangas, the main factor is food prices (79%), which lowers the risk. Jobs and employment come second everywhere, at 18% to 24%, and raise the risk. Fish kills had a small effect (1% to 2%) and raised the risk in Cavite, Laguna, Quezon, Rizal, and Batangas. Remittances from OFWs had a small effect (1%) and lowered the risk in Cavite, Laguna, Quezon, Rizal, and Batangas. Weather and climate had a small effect (1%) and raised the risk in Batangas. Of everything behind the CALABARZON index, about 12% comes from the regional hunger survey and the last published result, about 2% from the tone of the news and news reports that match none of the five factors, and about 86% from the five factors shown in the figure.

Figure 2

Factors Behind the Risk Across the Provinces, 2027 Q1



Note. Shares are combined across the five provinces on the scale of the regional index. Red bars raise the risk and green bars lower it; factors with no noticeable effect are not shown.

Table 2

Share of Each Factor by Province, 2027 Q1

Province	Food prices	Jobs and employment	Fish kills	Remittances from OFWs
Cavite	73% raises	24% raises	2% raises	1% lowers
Laguna	73% raises	24% raises	2% raises	1% lowers
Quezon	73% raises	24% raises	2% raises	1% lowers
Rizal	73% raises	24% raises	2% raises	1% lowers
Batangas	79% lowers	18% raises	1% raises	1% lowers

Note. Each row adds up to about 100% of the five factors. 'Raises' and 'lowers' show whether the factor pushes the province's risk score up or down.

Province Profile

Table 3 gives basic facts about each province. Cavite has the largest population (4,573,884), and Quezon has the highest poverty incidence (20%).

Table 3

Population and Poverty Incidence by Province

Province	Population	Poverty incidence
Cavite	4,573,884	9.9%
Laguna	3,687,345	6.0%
Quezon	2,260,669	19.7%
Rizal	3,416,541	7.1%
Batangas	2,994,795	8.8%

Note. Population is the total of the province's cities and municipalities in the PSA 2024 Census of Population (POPCEN). Poverty incidence is the share of people below the poverty line: Population-weighted mean of the PSA 2021 city and municipal Small Area Estimates of poverty incidence (2024 POPCEN weights), over the cities and municipalities with an estimate; Lucena City has none (highly urbanized city, not in the PSA no-HUC table). Derived by aiPHeed; not an official PSA province figure.

News Reports

No validated news reports about food conditions in the five provinces were published in 2027 Q1.

The tone of the news and news reports that match none of the five factors account for about 2% of everything behind the score; news matched to a factor is counted within that factor.

The news for 2027 Q1 is not complete: the quarter has not started yet. The 2027 Q1 forecast was built from the news of 2026 Q3: nine validated news reports.

Important Notes

- The risk score is the estimated chance that the combined hunger-and-food-price score is above its usual level. It is not a measure of hunger or of how many families lack food.
- By the study's rule (at least 75% correct on past quarters), results one quarter ahead are the reliable horizon: 78% were correct on development data. Results two quarters ahead were correct 67% of the time, so the two-quarter-ahead value shown here is an experimental preview.
- Results for 2023 Q1 to 2026 Q3 are backtests: each was estimated with only the data available before that quarter, but these quarters were also used while the model was being developed.
- No estimate for 2025 Q2 and 2025 Q3: their SWS hunger rounds are recorded as supplementary, not as the approved (definition A) series, so these quarters have no approved label to estimate or score. This is missing data, not a low score.
- City and municipal figures follow their province's score. They are not separate estimates for each city or municipality.
- This is not an official food insecurity classification. The report presents findings only; it does not recommend actions and should be used alongside DA Region IV-A's own guidance.